

Join Kata

The goal of this kata is to practice small nuances of join.
We will use CTEs to provide the proper data for each case.
That allows creating small cases for quick testing and feedback.

For example

```
with
Animals as (select "Dog" as animal_type
union select "Cat"
union select "Lion")
Select *
from
Animals
;
```

1.

Complete the following query so you will get Italian dishes. We would like to create a dish for each combination of pasta and sauce. Complete the query so you will get 12 dishes.

```
With Pasta as (select "Fettuccine" as pasta_type
# Complete here
```

```
)
, Sauce as (select "Tomatoes" as pasta_type
# Complete here
```

```
)
Select *
from
# Complete here
;
```

After which famous mathematician the operation that you used is named?

2.

Complete the following query so you will get dishes and their ingredients.
What should you do to have an existing ingredient that will not be part of the results?

With Dishes as (select 1 as dish_id, "Tomatoes Fettuccine" as dish_name
Complete here

)
, Ingredients as (select 1 as dish_id, "Tomatoes" as ingredient_name
Complete here

)
Select *
from
Complete here
;

3.

Complete the following query so you will get dishes and their ingredients.
This time you should **also** get dishes with no ingredients.
Make sure that you will give a funny name to the ingredients lacking dishes.

With Dishes as (select 1 as dish_id, "Tomatoes Fettuccine" as dish_name
Complete here

)
, Ingredients as (select 1 as dish_id, "Tomatoes" as ingredient_name
Complete here

)
Select *
from
Complete here
;

4.

Complete the following query so you will get dishes and their ingredients.
This time you should also get **only** dishes with no ingredients.
Make sure that you will give a funny name to the ingredients lacking dishes.

With Dishes as (select 1 as dish_id, "Tomatoes Fettuccine" as dish_name

Complete here

)

, Ingredients as (select 1 as dish_id, "Tomatoes" as ingredient_name

Complete here

)

Select *

from

Complete here

;

5.

Complete the following query so you will get dishes and their ingredients, just as in question 3. Put the tables in the opposite order that you did in question 2 yet get exactly the same result. What were the differences?

With Dishes as (select 1 as dish_id, "Tomatoes Fettuccine" as dish_name

Complete here

)

, Ingredients as (select 1 as dish_id, "Tomatoes" as ingredient_name

Complete here

)

Complete here

;

6.

Complete the following query so you will get dishes and their ingredients. This time you should also get **only dishes with tomatoes**. You can filter rows in two places. What is the difference?

With Dishes as (select 1 as dish_id, "Tomatoes Fettuccine" as dish_name

Complete here

)

, Ingredients as (select 1 as dish_id, "Tomatoes" as ingredient_name

Complete here

```
)  
Select *  
from  
# Complete here  
;
```

7.

Complete the following query so you will get dishes and their ingredients.
This time you should also get **only dishes without tomatoes**.

In principle, you can filter rows in two places. What is the difference? What is the difference from the previous question?

With Dishes as (select 1 as dish_id, "Tomatoes Fettuccine" as dish_name
Complete here

```
)  
, Ingredients as (select 1 as dish_id, "Tomatoes" as ingredient_name  
# Complete here
```

```
)  
Select *  
from  
# Complete here  
;
```

8.

Complete the following query so you will get dishes, their ingredients, and their cost.

With Dishes as (select 1 as dish_id, "Tomatoes Fettuccine" as dish_name
Complete here

```
)  
, Ingredients as (select 1 as dish_id, "Tomatoes" as ingredient_name  
# Complete here  
  
)
```

```
, Ingredients_cost as (select "Tomatoes" as ingredient_name, 5 as ingredient_cost  
# Complete here
```

```
)
```

```
Select *  
from  
# Complete here  
;
```

9.

Complete the following query so you will get dishes, their ingredients, and their cost.
This time select **only fancy dishes** that include an ingredient that cost at least 100.

```
With Dishes as (select 1 as dish_id, "Tomatoes Fettuccine" as dish_name  
# Complete here
```

```
)
```

```
, Ingredients as (select 1 as dish_id, "Tomatoes" as ingredient_name  
# Complete here
```

```
)
```

```
, Ingredients_cost as (select "Tomatoes" as ingredient_name, 5 as ingredient_cost  
# Complete here
```

```
)
```

```
Select *  
from  
# Complete here  
;
```

10.

Complete the following query so you will get dishes, their ingredients, and their cost.
This time select **no fancy dishes that** do not include an ingredient that cost at least 100.

Beware here and think how to use the answer to the previous question.
This question uses subqueries that appear later than join in the course.

Come back here after learning them.

What is the difference between this question and the previous one?

With Dishes as (select 1 as dish_id, "Tomatoes Fettuccine" as dish_name

Complete here

)

, Ingredients as (select 1 as dish_id, "Tomatoes" as ingredient_name

Complete here

)

, Ingredients_cost as (select "Tomatoes" as ingredient_name, 5 as ingredient_cost

Complete here

)

Select *

from

Complete here

;